Problem Chosen	2024	Team Control Number
B	**MCM/ICM**	**7391856**
	Summary Sheet	

Locating a Lost Submersible with a Recovery-Aware Readiness Plan

MCMS should combine independent acoustic tracking with a fault-dependent location model and a depth-compatible response contract. Our primary search heuristic targets a smaller restricted mean time to location. A separate recovery clock determines whether an early location can still be useful; it must not be confused with the probability of saving passengers.

Locate. A three-dimensional hybrid ensemble distinguishes sinking and seabed contact, suspension, and continued propulsion after communication failure. Persistent current uncertainty grows with elapsed time, while unresolved diffusion grows with its square root. At six hours, the conditional neutral-drift eastward standard deviation is 557 m; improving current information and the last fix reduces it to 236 m. Time-stamped pressure, motion, fault and life-support telemetry helps distinguish the search depths and keeps the last valid measurement's age explicit.

Prepare. Carry calibrated acoustic reception, a separately powered emergency beacon, sound-speed profiling and trained operators on the host. Compare ownership, long-term lease and response contracts by complete annual cost, availability and ready-at-scene delay. Under equal availability and intervention success, a six-hour rather than seven-hour compatible service justifies incremental annual cost only if $\Delta C/(\nu V) \leq 0.261729 a\rho$. Published survey/intervention examples reach 6,000/6,500 m, while a docking rescue system is limited to 610 m. These ratings have different purposes; deep human recovery needs its own engineered interface and timing evidence.

Search. In a specified synthetic deep-Ionian scenario, discovery-priority search attains 70.9% by six hours and 81.3% by eighteen hours, with restricted mean 6.456 h. A cumulative-coverage comparator using the same action domain and return reserve attains 70.2% and 7.726 h; a compact lawnmower attains 50.2% and 11.411 h. A recovery-priority candidate raises the reference timing/depth opportunity from 75.4% to 76.2%, at a cost of 2.35 minutes in restricted mean and 0.174 percentage points in final discovery. We retain discovery priority for the original search objective and present the recovery tradeoff as an option conditional on a credible extraction service. Delaying that service's arrival from six to seven hours reduces its discovery-priority opportunity to 49.2%; arrival at nine hours leaves zero opportunity under this clock.

Extrapolate. Replace currents, density, terrain barriers, launch limits and response access at a new destination. Unchanged nominal routes under stronger, rotated forcing attain only 47.3% final discovery. Multiple vehicles require distinct identities, shared-current inference, acoustic slots and joint dispatch. A six-assignment check gives 0.975 expected discoveries with 2.0 fleet-hours by searching different cells.

The numbers are conditional comparisons. Local trials must establish the acoustic likelihood, environmental errors and complete recovery service before authorizing an operating envelope. **Keywords:** hybrid drift; Bayesian search; persistent misses; readiness; recovery deadlines.

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1 The decision and the evidence available

The original problem asks MCMS to predict a lost submersible's location, select preparation equipment and costs, minimize time to location, calculate discovery probability after accumulated search results, and extend the plan to new destinations and multiple vehicles [1]. Location is the primary search endpoint. Passenger recovery adds compatibility and timing constraints to preparation; optimizing that endpoint would answer a different decision question. AI assisted formulation, computation and writing [9]; its actual role and record limits are disclosed after the solution.

No incident coordinates, hull parameters, measured currents, acoustic trials, prices or life-support duration are supplied. We therefore separate a transferable model from one numerical scenario. The model accepts archived currents, measured density and sound speed, real bathymetry and instrument likelihoods. Copernicus's Mediterranean reanalysis provides modeled currents, temperature and salinity on a 4–5 km grid with 141 levels [3]; this supplies a regional prior, rather than incident-scale certainty. TEOS-10 derives density and sound speed consistently from Absolute Salinity, temperature and pressure [2]. Local CTD and acoustic calibration remain necessary.

A spreading circle omits directional drift, depth modes and seabed stopping. Full fluid simulation would require unsupplied hull and boundary data. Our hybrid ensemble retains those decision-changing mechanisms without inventing a detailed hydrodynamic calibration. Its output is a joint distribution used by the search planner, not a claimed real-Ionian incident forecast.

2 Assumptions, reasons and their consequences

We use the following common assumptions to connect prediction, search and preparation. Local conditions are stated again only where they change a result.

1. **Regime uncertainty is retained.** The vehicle may sink, remain suspended or continue moving under propulsion. The scenario weights are conditional priors, because accident-frequency data are absent. An ascending branch is supported operationally, but the numerical deep case is conditioned on nonascending incidents; surfacing switches the search to radio/GNSS, visual and surface drift models.

2. **Current bias persists within an incident.** An uncertain forecast offset is sampled once, while small unresolved fluctuations enter diffusion. This separates parameter uncertainty from random motion. Repeatedly redrawing the bias would understate long-term dispersion.

3. **Seabed contact sticks in the principal scenario.** Contact halts a negatively buoyant vehicle if friction and terrain resist sliding. This is used to preserve bottom probability; its force criterion and a slipping alternative are explicit below.

4. **Detection is imperfect and persistent.** A correct-identification hazard acts only during an effective survey. Target quality is fixed, so poor reflectivity or shadow can cause cor-

related misses. Surveys do not clear cells with certainty. The box footprint and hazard are calibration hypotheses, not manufacturer specifications.

5. **Actions share physical resources and information.** All policies use the same preparation, travel, survey, depth, endurance and return rules and the same underlying ensemble. The planner knows a distribution, not the realized target. Timely observation summaries are required for a 15-minute update; unavailable summaries require a preplanned longer sortie.

6. **The recovery clock is a reference service.** The numerical opportunity assumes continuous tracking after discovery, fixed encounter depth for ascent, specified intervention times, zero extra observation delay and an already compatible recovery interface. These choices isolate the search/readiness tradeoff. They do not model waiting-time target motion, lifting dynamics, hull integrity or real intervention success.

The scenario in Table 1 specifies the comparison. Distributions without measurements are scenario inputs rather than estimated frequencies. A 120 s propagation step is retained; interpolation defines the trajectories on which the continuous search integrals are calculated.

Table 1. Conditional deep-Ionian scenario

Quantity	Specification
Last fix and seabed	$(0, 0, 2800)$ m; horizontal s.d. 100 m, depth s.d. 15 m; $H(x, y) = 3000 + 0.025x + 0.010y$ m
Current and diffusion	Mean $(0.050, 0.020)$ m/s; persistent-bias s.d. $(0.025, 0.015)$ m/s; $K = 0.2$ m^2/s
Fault regimes	Sinking/neutral/communication-only weights $(0.50, 0.40, 0.10)$; sinking speed $U[0.03, 0.12]$ m/s
Suspension and propulsion	Equilibrium-depth s.d. 35 m; relaxation 900 s; residual propulsion 0.06 m/s, heading s.d. $\pi/3$ rad
Movement and preparation	Host 5.0 m/s; sensor horizontal 1.8 m/s, vertical 0.5 m/s; preparation 0.5 h
Survey and detection	500 m square cell, depth half-band 100 m; 0.25 h survey, hazard $8q$ h^{-1}; $q \in \{0.4, 0.7, 0.9\}$ with equal weights
Resources and numerics	20 h wet endurance, 20% protected reserve; 2,400 trajectories, 120 s propagation, seed 7391861, horizon 18 h

3 Locate: buoyancy, drift and useful telemetry

3.1 Vertical regimes and seabed contact

Let $\mathbf{x} = (x, y)$ be east/north displacement in metres, $z \geq 0$ depth and $H(\mathbf{x})$ water depth. Before contact, the force balance is

$$m_{\text{eff}}\dot{w} = g\{m - \rho(S_A, \Theta, p)V(p)\} - \tfrac{1}{2}\rho C_D A_z w|w|, \qquad \dot{z} = w. \tag{1}$$

m_{eff} includes added mass; V is displaced volume, A_z vertical projected area and C_D drag coefficient. Each force has units N. Leakage, ballast and pressure-dependent volume change buoyancy. Under constant conditions the terminal vertical speed is

$$w_\infty = \operatorname{sgn}(m - \rho V)\sqrt{\frac{2g|m - \rho V|}{\rho C_D A_z}}. \tag{2}$$

The scenario uses a sinking-speed range because the hull parameters are unknown. A neutral equilibrium requires $m = \rho V$; stable trim additionally requires $\partial_z[g(m - \rho V)] < 0$. Density equality alone does not ensure a fixed depth. Suspended trajectories use $\dot{z} = -(z - z_*)/(900\,\text{s})$ around an uncertain stable depth z_*. This is a trim surrogate, not a measured density law.

At $z = H(\mathbf{x})$, the principal contact transition stops vertical and horizontal motion. If $N > 0$ is submerged normal load, a sufficient sticking condition is

$$\|\mathbf{F}_{\text{drag},\parallel} + \mathbf{F}_{\text{gravity},\parallel}\| \le \mu N. \tag{3}$$

Near-neutral buoyancy makes N small and can invalidate sticking. A bottom-slip branch then follows the seabed tangent subject to friction and obstacles. Repeated bottom-position observations distinguish it. No trajectory may pass through seabed or land; reaching $z = 0$ switches to surface search. The comparison uses a sloping plane with no coast or cliff, so operational terrain must add those barriers.

3.2 Persistent current uncertainty and location modes

For a mobile vehicle,

$$d\mathbf{x} = \{\overline{\mathbf{u}}(\mathbf{x}, z, t) + \mathbf{b} + \mathbf{v}_R\}dt + \sqrt{2\mathbf{K}}\, d\mathbf{W}_t. \tag{4}$$

The archived current is $\overline{\mathbf{u}}$; $\mathbf{b}$ is a trajectory-specific fixed bias. Propulsion-loss states have $\mathbf{v}_R = 0$, while communication-only failure retains a heading distribution. $\mathbf{K}$ has units m^2/s. In the constant-current neutral limit,

$$E[\mathbf{x}(t)] = \mathbf{x}_0 + \overline{\mathbf{u}}t, \qquad \Sigma_x(t) = \Sigma_0 + t^2\Sigma_b + 2\mathbf{K}t. \tag{5}$$

Thus persistent current error grows quadratically in position variance. A six-hour eastward standard deviation is $\sqrt{100^2 + (0.025 \cdot 21600)^2 + 2(0.2)(21600)} = 557$ m. Reducing the current-bias s.d. to 0.010 m/s and last-fix s.d. to 25 m gives 236 m. In this case, measuring currents is more valuable than merely polishing the coordinate.

Particles retain joint position, regime and fixed detectability:

$$\pi_t(d\mathbf{x}, dz, dR, dq) = \sum_{i=1}^{N} p_i \delta_{(\mathbf{x}_i(t), z_i(t), R_i, q_i)}. \tag{6}$$

Figure 1 shows the practical consequence. Bottom probability stops, while suspended and propelled states keep spreading at different depths. A horizontal ellipse cannot determine the right survey depth. The existing ensemble's empirical 95% horizontal containment areas are 0.95, 8.2 and 73.4 km^2 at two, six and eighteen hours; these describe the conditional ensemble rather than calibrated confidence regions.

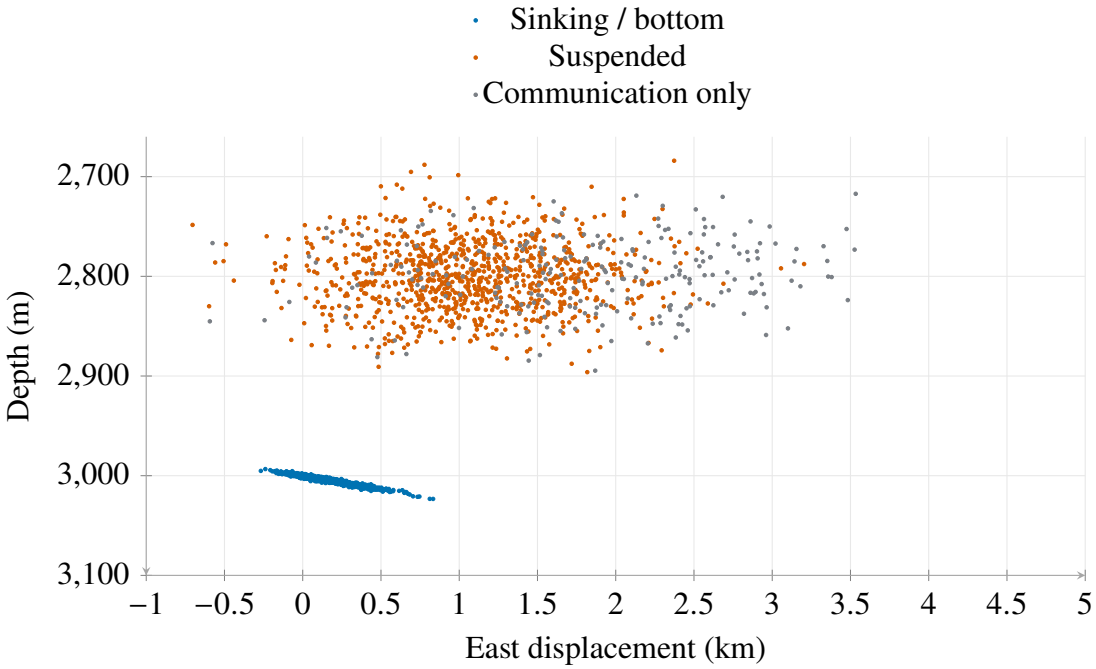


Figure 1. Fault-dependent location at six hours

3.3 What to send, how to interpret it and when to escalate

Each packet should identify the vehicle and measurement time, give position with covariance and sensor status, pressure depth and depth rate, velocity and reference frame, heading/attitude, thruster commands, ballast/leak alarms and remaining battery, oxygen and carbon-dioxide margins. Pressure and an altimeter help distinguish suspension from bottom contact. DVL/INS track motion between acoustic fixes; loss of bottom lock requires growing inertial uncertainty. CTD/sound-speed measurements help both buoyancy and acoustic ranging.

The host combines GNSS, attitude and calibrated USBL. Ranger 2 advertises 11 km tracking, optimized accuracy of 0.04% of slant range and two-way telemetry [4]. These are conditional maxima, not achieved accuracy for MCMS. A protected emergency pinger needs a separate battery and activation path to avoid a common power failure. Correlated calibration errors in USBL and DVL/INS must not be fused as independent measurements. With trusted observation y, reweight by $p_i^+ \propto p_i^- L(y \mid X_i, R_i)$, including sensor-fault likelihoods.

At declaration, use last-valid-fix age $A = t_{\text{declare}} - t_{\text{last fix}}$. With interval Δ and processing delay ℓ, two missed reports give $A = 2\Delta + \ell$, rather than Δ. A delayed received fix is aged from its measurement timestamp. Accept a cadence/trigger only if the predicted containment radius $r_{95}(A)$ remains within a tested emergency-acquisition radius and declaration leaves a safe depth/ballast response margin. This is a site-dependent rule.

An illustrative sinking-status bit with likelihoods $(0.90, 0.10, 0.15)$ and the stated fault prior has positive-report probability 0.505. Its positive/negative sinking posteriors are 0.891/0.101, and expected regime entropy falls from 1.361 to 0.844 bits. This establishes how telemetry can separate modes; an actual time benefit requires measured likelihoods and the subsequent dispatch comparison.

4 Prepare: capability, cost and the response clock

4.1 A layered equipment plan

The host should carry calibrated acoustic tracking/telemetry and independent emergency reception, spare beacon batteries/transducer, CTD or sound-speed profiling, navigation logs, safe lifting arrangements and trained watchkeepers. Its role is rapid reacquisition before uncertainty spreads. A shallow inspection ROV is useful only within its rated depth.

The deep layer needs a search vehicle with bottom imaging and tested midwater detection, terrain avoidance, navigation aiding, adequate duty-cycle endurance, launch/recovery hardware and prompt target/miss summaries. HUGIN 6000 provides a 6,000 m platform example, with 2–5 knot speed and an advertised 240 m synthetic-aperture sonar range at 3 knots [5]. A bottom-imaging swath is not a midwater detection volume. Our effective 500 m cell must be calibrated across altitude, attitude, slopes and acoustic conditions. Raw imagery retrieved after a long sortie cannot support an immediate 15-minute replan.

A response vessel adds depth-rated intervention imaging, manipulators, compatible tooling, station keeping, winch/crane/cable, navigational support, trained crews and medical capability. Jason reaches 6,500 m and carries cameras, sonar and manipulators [6]; this does not establish lifting a human submersible. Such recovery needs certified lift points, rigging, load and buoyancy estimates, seabed clearance, controlled ascent and surface handling. The NATO rescue vehicle docks at a compatible hatch only to 610 m and carries up to 15 passengers with transfer under pressure [7]. It cannot serve the 3,000 m case simply because it is a rescue system. MCMS must demonstrate an engineered emergency ascent or a compatible extraction method within life support; otherwise reduce the authorized operating depth.

4.2 Annual cost and access versus ownership

For package e, annual equivalent cost is

$$\begin{aligned} C_e = {} & \mathrm{CRF}(i,n)P_e + M_e + T_e + R_e \\ & + \nu\{C_{\text{mobilize,e}} + c_{\text{ship,e}}h_{\text{ship}} + c_{\text{use,e}}h_{\text{use}}\}, \end{aligned} \tag{7}$$

where P_e is capital, M_e servicing/consumables, T_e training/calibration, R_e standby retainer and ν annual activation frequency. $\mathrm{CRF} = i(1+i)^n/\{(1+i)^n - 1\}$ annualizes capital. Quotes must include depth options, battery cycles, operators, vessel compatibility, transport and emergency commitment. A historical UK contract extension exceeded £7 million for three years of NSRS support, maintenance, upgrades and readiness [8]; it is evidence of readiness obligations, not a current MCMS quotation.

For a simplified constant extraction allowance, let D be the latest useful discovery time. An additional package is preferred economically when

$$\Delta C \le \nu V\{a_e F_e(D; d_e) - a_0 F_0(D; d_0)\}, \tag{8}$$

where a_e is tested availability, d_e access delay and V an explicit consequence valuation. $F_e(D; d_e)$ must propagate the target during delay and evaluate the actual delayed policy; shifting a moving-target curve by d_e would reset the incident clock incorrectly. When extraction

depends on depth/regime, use the integrated opportunity below, then measured intervention success. Safety requirements can impose a readiness floor independent of monetization.

Carry or secure the rapid acoustic package before every dive; this requirement does not determine ownership. Compare a purchased package, a long-term onboard lease and a response contract after requiring equal depth, sensing and recovery capability. Table 2 identifies the inputs needed for that choice. Count bundled costs once. No current quotation or serviceability probability is supplied.

Table 2. Acquisition choices and the inputs to compare

Input	Purchase	Long-term lease	Response contract
Capital / rental	Annualized P_e	Annual lease fee	Annual retainer
Maintenance / training	MCMS pays $M_e + T_e$	Quote states included service and crew training	Contract states servicing and trained crew commitment
Standby	Ready crew, deck and spares	Equipment kept onboard with agreed standby	Guaranteed call-out coverage and fallback
Mobilize / use	Own launch, ship and duty costs	Delivery, launch, usage and support fees	Mobilization, voyage, launch and per-use fees
Availability	Tested a_e, unknown here	Tested leased-system a_e, unknown here	Contracted and tested a_e, unknown here
Scene-ready time	Actual crew/launch readiness	Same onboard target if maintained	Full call-out, voyage and operational setup d_e

For alternatives with equal conditional search/recovery capability, define useful service $J_e = a_e \rho_e Q_e(d_e)$, where ρ_e is a constant conditional intervention-success factor for this example. This factorization is an explicit comparison assumption; if success depends on state/time, use the integral in Equation (11) instead. The economic rule is $\Delta C/(\nu V) \leq J_e - J_0$, with measured safety/readiness requirements enforced separately.

The existing fixed-route service scenarios give a concrete threshold without inventing prices:

$$\frac{\Delta C}{\nu V} \leq 0.753606 a_6 \rho_6 - 0.491877 a_7 \rho_7. \tag{9}$$

With equal $a\rho$, this is $0.261729 a\rho$. Illustrative $a = 0.9$, $\rho = 0.8$ give 0.188445; these are declared assumptions, not estimated equipment performance. If ownership and onboard leasing have equal availability, success and ready time, their modeled benefit is equal: choose the lower complete annual cost, subject to a reliable term and crew arrangement. A faster purchase is worth its premium only when its measured readiness advantage satisfies Equation (9). Otherwise use the cheaper qualified lease/contract. A seven-hour contract's lower price cannot compensate for a mandatory six-hour safety requirement. MCMS must obtain quotations and readiness evidence to place its alternatives on this decision boundary.

4.3 A reference recovery service and its latest useful discoveries

Let t be the first correctly identified discovery, δ additional observation/confirmation latency, d_r recovery-service readiness at the operation, I_R the regime-dependent target approach/intervention duration, and h_s surface handling. With encounter depth $z(t)$ and ascent speed v_a in m/s, the reference completion time is

$$c(t, X, R) = \max(t + \delta, d_r) + I_R + \frac{z(t)}{3600 v_a} + h_s. \tag{10}$$

All other times are hours. The service assumes continuous target tracking and uses the discovery depth throughout the modeled ascent allowance. Waiting-time drift or depth changes, towing/lifting dynamics and confirmation failures require additional engineering. I_R and d_r must include their declared work; no readiness or travel time becomes free on discovery. Our effective hazard below already represents correct identification, so the scenario has $\delta = 0$.

Define $b_i(t) = 1$ when this service is depth-compatible and $c(t, X_i, R_i) \le L - r$, with life support L and reserve r. The reference useful-discovery opportunity is

$$Q_\pi = \sum_i \int_0^H s_i(t) \lambda_i(t) b_i(t)\, dt. \tag{11}$$

The true probability under the same reference service also multiplies the integrand by unmeasured conditional availability/intervention success $\rho_i(t) \in [0, 1]$, hence it is at most Q_π. This upper bound is relative to that clock and compatibility model; it is not universal across different recovery technologies or moving-depth models.

We retain $L = 18$ h, $r = 2$ h, $d_r = 6$ h, $I_R = (3, 5, 4)$ h, $v_a = 0.25$ m/s, $h_s = 1$ h and an already-compatible 6,000 m package. At 3,000 m ascent alone takes $10/3$ h. A suspended target at 2,800 m needs discovery and service readiness by approximately $16 - 5 - 2800/900 - 1 = 6.89$ h; a 3,000 m sinking/bottom target permits approximately 8.67 h. This explains why a late bottom discovery may remain useful when a shallower suspended discovery does not. The 610 m docking tool has zero depth eligibility in this scenario. None of these timing assumptions estimates an actual passenger survival curve.

5 Search: use remaining probability without clearing missed cells

5.1 Detection, misses and discovery probability

An action a covers a 500 m square and a depth half-band of 100 m. Let $K_a(X_i(t))$ indicate this footprint. During survey its correct-identification hazard is $\lambda_i(t) = 8 q_i K_a(X_i(t))$ h^{-1}; during transit it is zero. Quality q_i is fixed throughout the incident. With initial $s_i(0) = p_i$, a miss gives

$$s_i^+ = s_i^- e^{-8 q_i E_{ia}}, \quad E_{ia} = \int_{t_a}^{t_a + \tau_a} K_a(X_i(u))\, du, \quad p_i^+ = \frac{s_i^+}{\sum_j s_j^+}. \tag{12}$$

If A/B masses are 0.6/0.4 and a pass identifies A with probability 0.8, a miss leaves total survival 0.52 and A posterior $3/13$. The next A pass offers $(3/13)0.8$ conditional chance, rather

than the original 0.48. Correct identification is essential: unconfirmed contacts require an inspection/data-association task and cannot be counted as first location.

For a policy's all-negative branch,

$$S_\pi(t) = \sum_i s_i(t), \quad F_\pi(t) = 1 - S_\pi(t), \quad M_\pi = E[\min(T, H)] = \int_0^H S_\pi(t)\, dt. \tag{13}$$

Restricted mean remains finite even if a target is never found. Before first discovery the only search outcomes are misses, so this branch determines the next action and its entire first-discovery distribution. It can be precompiled; route quality is not a separate advantage of online computation. New environmental measurements, fault reports, equipment status or nonterminal contacts create additional information branches and justify replanning.

After no discovery through t, the continuation probability is

$$P(T \le t + u \mid T > t) = 1 - \frac{S_\pi(t+u)}{S_\pi(t)}. \tag{14}$$

The original nominal calculation gives about 31.3% for discovery by twelve hours after misses through six hours. Past effort, last-fix age and used endurance do not reset when new data arrive.

5.2 Feasible deployment and the discovery-priority route

The candidate lattice is $x \in [-2, 8]$ km, $y \in [-4, 5]$ km in 500 m steps, at bottom-survey depth $H(x, y) - 40$ m or suspended depth 2,800 m: 798 actions. The first common deployment is $(0, 0, 2960)$ m. It is a fixed comparison convention supported by the last-fix-centered modal bottom mass and the sinking prior of 0.5; it is not a proven best initial point over all 798 actions. Preparation and descent put its first survey at 2.144 h. Later horizontal/vertical movement is charged conservatively as

$$d_a = \frac{\|\mathbf{x}_a - \mathbf{x}_{\text{sensor}}\|}{3600 v_h} + \frac{|z_a - z_{\text{sensor}}|}{3600 v_z}, \qquad t_a = t + d_a. \tag{15}$$

Speeds here are m/s; initial horizontal travel uses the host, subsequent travel the sensor. A survey lasts 0.25 h. With launch time t_l and actual launch position $\mathbf{x}_l$, require

$$t_a + \tau_a + d_{\text{return}}(a, \mathbf{x}_l) \le t_l + 16 \text{ h}. \tag{16}$$

This protects 20% of 20 h wet duty-cycle endurance and charges descent plus actual return. It is an endurance proxy needing duty-cycle trials, not a full energy model.

The primary route approximates minimization of M_π by future-arrival gain per elapsed time:

$$G_a = \sum_i s_i(t)\{1 - e^{-8 q_i E_{ia}}\}, \qquad a^* = \arg \max_{a \text{ feasible}} \frac{G_a}{d_a + \tau_a}. \tag{17}$$

An 80-action shortlist uses posterior mass at $t + 0.5$ h weighted by q for screening; every candidate is then checked at its own arrival. Five midpoint samples evaluate proposed exposure. Selected routes are evaluated separately by continuous event integration. The heuristic allows revisits and follows moving probability; it is not a global optimum of the full routing problem. Probability outside the action domain remains in $S(t)$.

For disjoint one-visit cells without travel/drift, completion-time identification probability $p_j d_j$ and duration τ_j, a descending $p_j d_j/\tau_j$ order minimizes the restricted mean at $H = \sum_j \tau_j$. An adjacent swap changes the objective by $p_j d_j \tau_i - p_i d_i \tau_j$, proving this special-case order. Motion, travel and repeat searches break that separability; Equation (17) is motivated by the special case, rather than inheriting its global guarantee.

5.3 The realized route and its reproduction

Figure 2 exposes the recommended horizontal route. Figure 3 exposes the layers and time of its 34-survey all-negative plan. The map connects survey centers in order; overlapping centers represent revisits, not extra uncharged observations. Bottom squares and suspended circles mark different survey layers. Table 3 gives the first six visits and final visit, in metres relative to the last fix and hours since that fix. Launch occurs at 0.500 h at $(0, 0, 0)$ m; after survey 34 ends at 14.417 h, the sensor returns to that launch point at 16.358 h. The depth plot connects transit endpoints schematically: horizontal motion is charged in the orthogonal travel rule, while the colored horizontal segments alone are surveys.

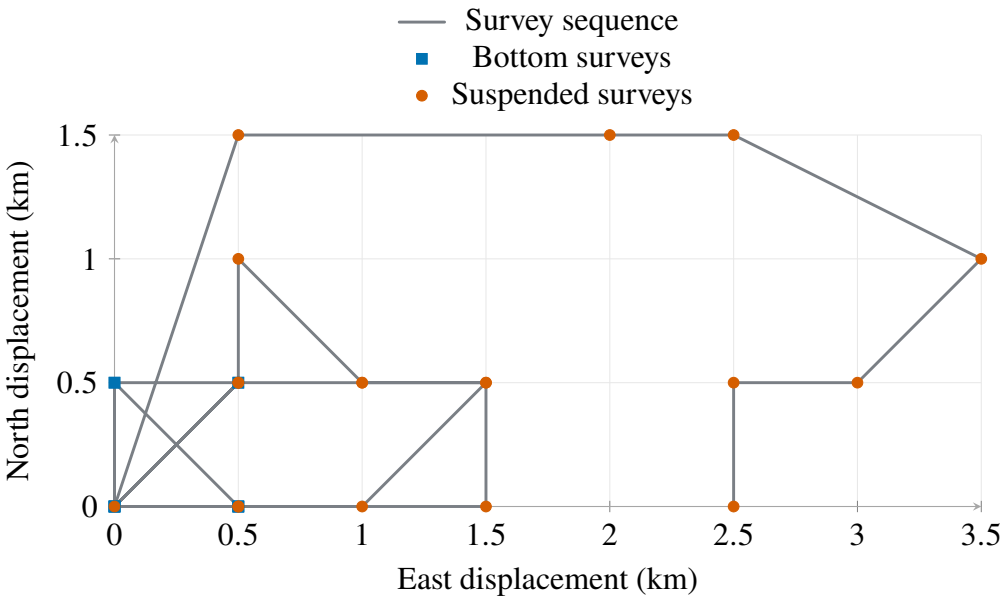


Figure 2. Recommended horizontal survey sequence

The implemented discovery-priority procedure is:

1. Load the scenario and joint trajectories; initialize surviving weights $s_i = p_i$, the preparation clock and unused endurance.

2. Apply the declared common first action; subsequently shortlist 80 cells from surviving mass at $t + 0.5$ h.

3. For each candidate, propagate to its own arrival, estimate survey exposure, reject depth/time/return violations, and score $G_a/(d_a + \tau_a)$.

4. Execute the highest-scoring feasible action; record coordinates, layer, transit and survey times; multiply weights by the negative-survey likelihood.

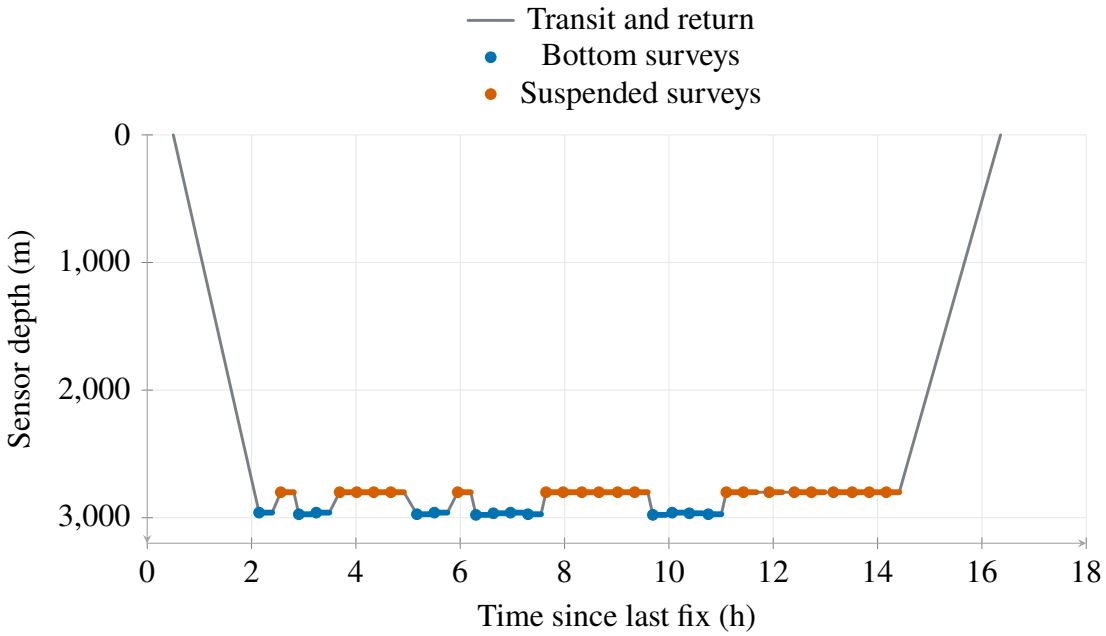


Figure 3. Survey layers and the complete sortie clock

Table 3. Initial, final and return actions

Visit	East (m)	North (m)	Depth (m)	Start (h)	End (h)
1	0	0	2960.0	2.144	2.394
2	500	0	2800.0	2.560	2.810
3	500	0	2972.5	2.906	3.156
4	0	0	2960.0	3.240	3.490
5	500	500	2800.0	3.688	3.938
6	1000	500	2800.0	4.016	4.266
34	2500	0	2800.0	14.167	14.417
Return	0	0	0	14.417	16.358

5. Continue after actual misses without resetting the clock; stop when no feasible next survey remains and return to the actual launch point.

The supplied companion index gives the exact inputs, action object, interpreter/workflow commands and hashes. Table 4 is its entry map; paths are relative to the delivered case. Existing numerical evidence is reproduced rather than silently regenerated outside its remaining-budget rule. The run's `output/results.json` contains all 34 discovery-priority actions; the frozen seed and coverage results are indexed alongside the ensemble. The public inherited 22-page demonstration remains available in the fixed Praxis demo collection. This private successor's complete case and reproduction index are supplied locally, without claiming a new public URL.

5.4 References with cumulative effort and a common probability evaluator

The compact lawnmower starts in the same modal bottom cell, then covers complete rows of $x = -500, 0, 500, 1000, 1500$ m and $y = -500, 0, 500, 1000$ m, alternating bottom and sus-

Table 4. Companion reproduction entry map

Object	Actual entry
Index	`reproduction-index.json`
Frozen location arrays	`raw/data/ensemble.npz`
Scenario parameters	`code/config.json`
Selected action object	`routes/discovery` in run results
Successful run	`20261013T015749-bd287c32`
Model / independent check	`code/model.py`; `code/validate.py`
Continuous metrics	Run output `independent-results.json`

pension layers. It is feasible in the common domain, while its compact field is an explicit policy choice.

The stronger comparator shares all 798 candidates, future-arrival evaluation, preparation, movement, survey and return rules. It remembers trajectory geometric exposure C_i and scores

$$G_a^{\mathrm{cov}} = \sum_i p_i \frac{\min\{E_{ia}, (\tau - C_i)_+\}}{\tau}, \qquad C_i \leftarrow C_i + E_{ia}, \quad \tau = 0.25 \text{ h}. \tag{18}$$

Its shortlist uses remaining geometric coverage without q weighting. After one full survey's exposure, a trajectory receives no more coverage credit; probability evaluation still uses the same persistent q and Equation (12). Thus full geometric exposure is not certain discovery: at $q = 0.4$, 0.25 h exposure identifies only $1 - e^{-0.8} = 55.1\%$. The comparator tests a distinct route construction, not all possible systematic planners.

6 When earlier location and useful recovery prefer different actions

6.1 An exact counterexample and a candidate search rule

The original location objective need not optimize Q. Consider three disjoint static cells A/B/C, perfect identification at survey completion, no travel, one resource and one-hour surveys. Prior masses are $(0.5, 0.3, 0.2)$ and latest eligible discovery times $(3, 1.5, 2.5)$ h. All six orders consume three hours and eventually discover with probability one.

The location-optimal A/B/C order has $M = 0.5(1) + 0.3(2) + 0.2(3) = 1.7$ h and $Q = 0.5$. The B/C/A order has $M = 2.2$ h and $Q = 1$. It finds B and C while they are still eligible, then A before its later deadline. Exhaustive rational enumeration gives the unique minimum mean and maximum opportunity in this six-order domain. This is a legal scheduling counterexample, not a calibrated miniature ocean incident. It disproves the implication that minimizing location time always maximizes recovery opportunity.

For the full domain, a recovery-priority candidate replaces G_a by

$$G_a^Q = \sum_i \int_{t_a}^{t_a+\tau_a} s_i(t) \exp\left[- \int_{t_a}^{u} \lambda_i(v)\, dv \right] \lambda_i(u) b_i(u)\, du. \tag{19}$$

Use $G_a^Q/(d_a+\tau_a)$ for selection and timely-eligible surviving mass for screening, with the same physical domain and initial survey. If every feasible useful gain is zero, revert to the discovery score and shortlist so late location value is retained. This is a local heuristic with an explicit endpoint; it does not prove maximum Q over all routes.

6.2 The same-resource comparison and the choice it supports

Table 5 reports continuous first-event integration on identical stored nominal trajectories and a common independently drawn stronger/rotated-current ensemble. Every nominal route is replayed unchanged in the challenge, preserving information conditions. Q uses the reference service in Section 4.

Table 5. Location and recovery opportunity under a common clock

Environment / route	$F(6)$ (‰)	$F(18)$ (‰)	M (h)	Q (‰)
Nominal / Discovery priority	70.9	81.3	6.456	75.36
Nominal / Recovery priority	70.9	81.1	6.495	76.18
Nominal / Cumulative coverage	65.7	70.2	7.726	67.34
Nominal / Compact lawnmower	26.1	50.2	11.411	41.36
Challenge / Discovery priority	43.3	47.3	11.195	45.64
Challenge / Recovery priority	43.3	46.9	11.209	46.39
Challenge / Cumulative coverage	36.5	40.3	12.153	36.86
Challenge / Compact lawnmower	23.9	46.0	11.963	43.70

The original discovery-priority route remains the recommendation for minimizing time to location among these candidates. In the nominal case it improves final discovery by 11.08 percentage points and restricted mean by 1.270 h relative to cumulative coverage. Figure 4 shows the timing gain, including flat periods when no survey occurs. The larger difference against the compact lawnmower includes following drift beyond its field. All four policies' probability evaluators retain cumulative misses.

Recovery priority gains 0.823 percentage points of opportunity, at 2.35 minutes of additional restricted mean and 0.174 points less final discovery. The opportunity gain is entirely in the sinking/bottom branch: 47.42% to 48.24% of total prior mass; other branches are unchanged. It therefore matters only if the response service can actually perform bottom recovery. Figure 5 shows the extra useful discoveries as a cumulative difference between the two routes. The candidate redirects some later effort before that branch's deadline, rather than increasing early discovery through six hours.

Discovery and recovery priority are nondominated in $(M, -Q)$ within these four nominal candidates; cumulative coverage and the compact lawnmower are dominated by discovery priority in these two metrics. This is a finite comparison, not the full Pareto frontier of every feasible route. MCMS should retain the faster location plan unless a demonstrated recovery requirement justifies the small delay. If conditional intervention success varies with time, position or regime, recompute the success-weighted objective; ordering by an unmeasured success function would manufacture information.

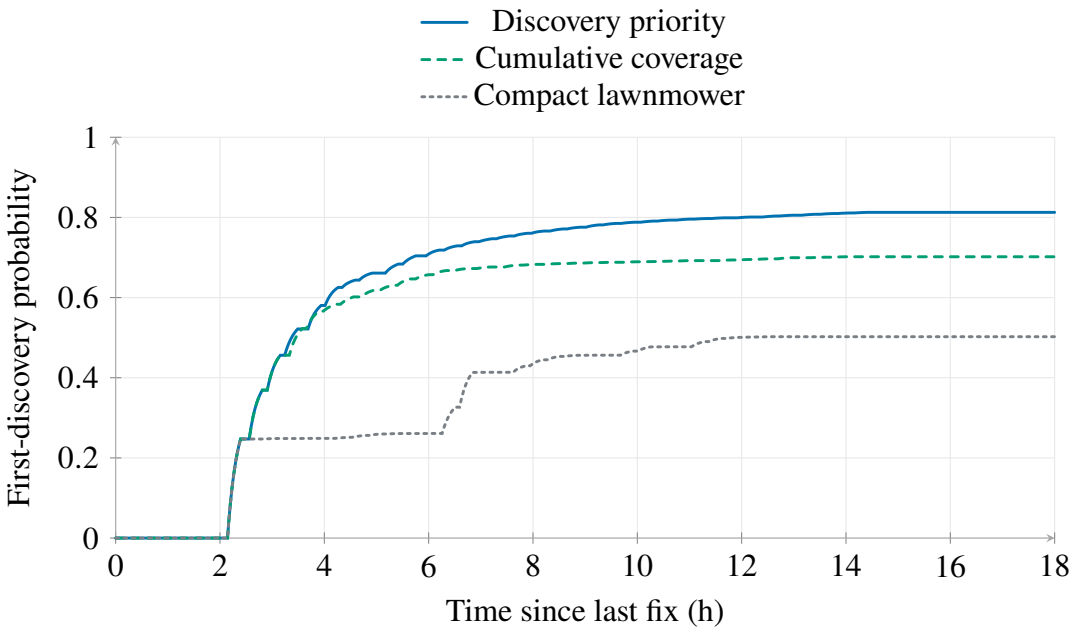


Figure 4. Discovery under nominal forcing

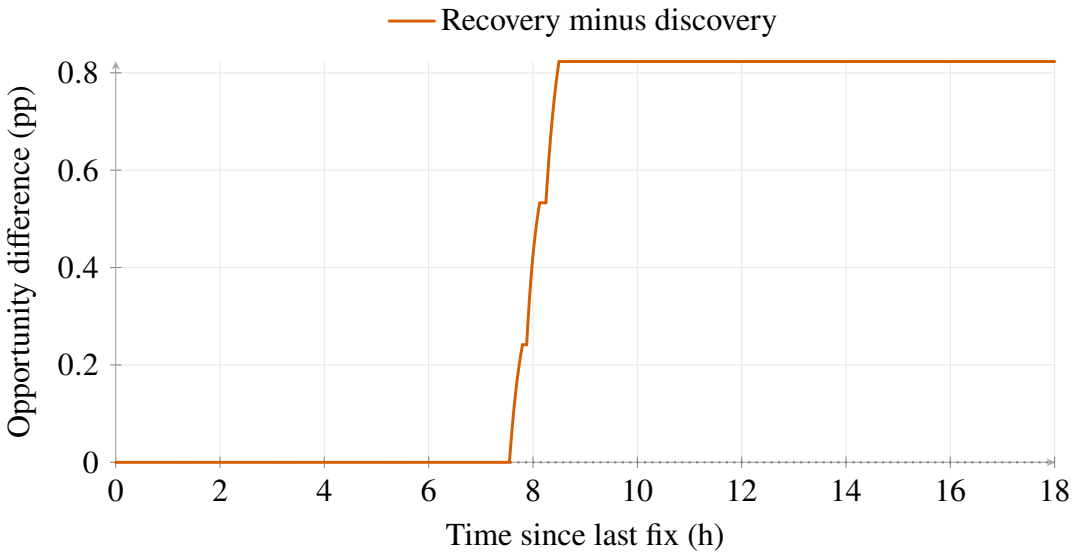


Figure 5. Recovery minus discovery opportunity

All sorties preserve the return reserve. Usable wet times are 15.858/15.822 h for discovery/recovery priority, with minimum return margins 0.142/0.178 h; their return arrivals are 16.358/16.322 h after the last fix. Preparation is outside wet endurance but inside the incident clock. The comparison uses common resource limits, not equal realized expenditure or equal survey count for every heuristic.

6.3 Readiness can matter more than the route change

Table 6 keeps both nominal routes and all detection assumptions fixed, changing only recovery-service readiness. Early arrival at four rather than six hours gives no extra opportunity under this service because both precede the useful completion limits. Arrival at seven hours removes suspended eligibility; at eight hours it also removes the communication-only branch. At nine hours no discovery can fit the reference service. This is a scenario threshold, not an empirically estimated rescue-arrival distribution.

Table 6. Fixed routes with different service readiness

Ready at (h)	Discovery priority Q (%)	Recovery priority Q (%)
4	75.36	76.18
6	75.36	76.18
7	49.19	50.01
8	47.42	48.24
9	0.00	0.00

Reducing service delay across an eligibility threshold is more consequential here than the 0.823-point route change. The finding strengthens the preparation recommendation: a contract must specify arrival at a ready, compatible operating state, rather than a promise to start mobilizing. Where that state cannot be reached before the measured life-support margin, better sonar alone cannot qualify the operation.

7 Verification, model risk and the escalation boundary

7.1 Independent calculations on the declared model

The route evaluator uses 100 midpoint detector samples per survey. A separate validator imports neither the planner nor its geometry module and uses no expected route answers. It reconstructs movement, each return allowance, lattice/depth membership and every linear-trajectory entry/exit from stored coordinates. Within each intersection, it integrates exponential survival and restricted mean analytically, then splits at recovery-readiness and deadline crossings. Maximum midpoint/event differences are below 0.002 percentage points for final discovery, 0.002 points for opportunity and 0.00024 h for restricted mean across the eight route/environment pairs. Thus the displayed comparative precision is supported on the stored trajectories.

The same integrator is tested against a stationary single target with hazard 2/h over a 0.25

h survey. Its discovery is $1 - e^{-0.5}$ and its restricted mean is the sum of the pre-survey delay, $(1-e^{-0.5})/2$, and the remaining horizon weighted by $e^{-0.5}$. A service arriving at 20 h gives zero opportunity without changing discovery; a 610 m depth limit also gives zero opportunity for the deep target. Exact rational enumeration checks the six-order counterexample and the $10/3$ h ascent conversion. Surface/seabed and all route-return constraints are checked separately from probability arithmetic.

These are calculation checks under piecewise-linear 120 s trajectories. They do not estimate incident currents, verify the hull's motion or validate the detector and recovery service. Existing same-model sampling checks replayed the original nominal routes on 9,600 independent particles and gave about 80.3%/50.8% final discovery for discovery-priority/lawnmower, versus the original 81.2%/50.2% detector-quadrature values. The new recovery candidate has not been tuned across seeds; its small opportunity increment should be treated as conditional, with local measurement uncertainty larger than its printed precision.

7.2 Stronger forcing and persistent detection failure

The forcing challenge changes current mean to $(0.100, -0.025)$ m/s and bias s.d. to $(0.050, 0.035)$ m/s. All four nominal routes face the same generated challenge trajectories. Figure 6 shows discovery priority falling to 47.3%, close to the lawnmower's 46.0%. Recovery priority adds 0.752 opportunity points while increasing restricted mean by 0.81 minutes. The narrow finite-scenario tradeoff persists, but neither route protects against an incorrect environmental field.

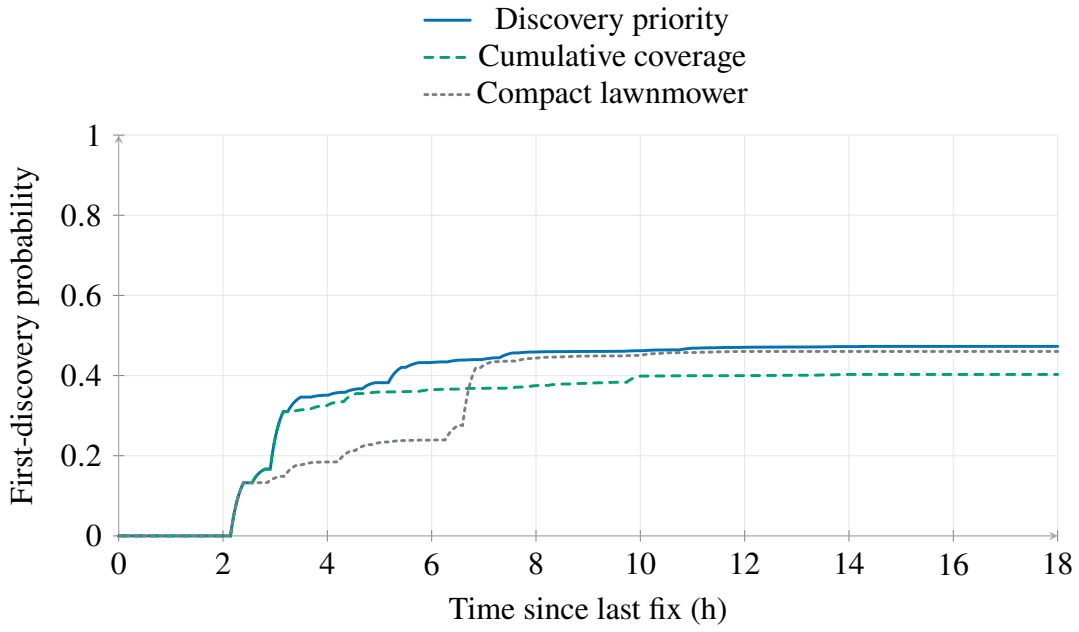


Figure 6. Discovery under stronger, rotated forcing

Persistent invisibility is a different risk. If a sensor cannot detect prior mass η, then $S(t) \geq \eta$ and $F(t) \leq 1 - \eta$ no matter how often it repeats. With 10% invisible mass and 90% per-pass identification probability 0.8, two passes leave $0.1 + 0.9(0.2)^2 = 0.136$ survival; replacing this by independent per-pass mean probability 0.72 gives only $0.28^2 = 0.0784$. That exact 5.76-point

false-confidence gap motivates persistent quality in the posterior and an independent modality after difficult misses.

The retained model-risk experiment used an 8% invisible prior state and 15% alternative forcing component. Its reoptimized route reached about 69.5% final discovery, and did not improve restricted mean relative to replaying the nominal route on that same mixture. We retain this failure as a limit of gain-per-time construction. It is not evidence that extra optimization always helps. Improved-current, changed fault-prior and bottom-slip comparisons identify current information, regime classification and bottom behavior as measurement priorities; their conditional performance does not establish a continuous-route optimum.

After repeated misses, obtain current/CTD information, reassess slipping or residual propulsion, check depth bands and acoustic shadow, and add an independent sensor if warranted. A miss should update both location and detectability. It should not force confidence into a field that contradicts new observations. A new observation is incorporated at its actual receipt time, from the sensor's real location and remaining endurance.

8 Extrapolate: regional forcing and multiple identities

8.1 New destinations

The equations transfer to the Caribbean; the scenario coefficients do not. Replace depth-resolved current forecasts and observations, temperature/salinity and sound speed, coastal/reef bathymetry, tide/wind forcing for surface states, sea-state launch limits, port access and recovery commitments. Paths must respect land and terrain barriers. The stronger/rotated-current comparison demonstrates a possible consequence of forcing mismatch, not a measured Caribbean prediction. A new site requires a requalified operating envelope and local trials.

8.2 Shared currents, acoustic slots and joint dispatch

For submersibles $k = 1, \ldots, m$, maintain labeled distributions $\pi_t^{(k)}$ and unique identifiers. Shared current errors couple their locations: safe vehicles can inform common forcing, while treating their errors as independent would overstate information. An uncertain-identity contact needs a data-association likelihood and cannot count as locating all targets.

At 3,000 m and nominal sound speed 1,500 m/s, acoustic round-trip time alone is about four seconds. Three sequential exchanges require at least twelve seconds plus processing/guards; a one-second aggregate polling promise cannot satisfy that schedule. Codes, slots, guard bands and collision-safe spacing protect the assumed observations.

For resource e, target k and action a, use the target's surviving probability to form gain g_{eka} and task duration τ_{eka}. Assign one simultaneous task per resource, compatible depth/tooling, nonconflicting acoustic cells and travel/time windows. For fixed fleet effort maximize $\sum g_{eka}x_{eka}$; gain per total effort is a separate ratio objective. A six-assignment check for two targets and shared cells A/B/C uses masses $(0.60, 0.25, 0.15)$ and $(0.10, 0.35, 0.55)$, detection rows $(0.8, 0.7, 0.5)$ and $(0.7, 0.8, 0.9)$, and effort rows $(1.0, 1.5, 2.0)$ and $(2.0, 1.2, 1.0)$ h. Excluding a shared simultaneous cell selects A/C: $0.60(0.8) + 0.55(0.9) = 0.975$ expected

discoveries in 2.0 fleet-hours. The next best A/B gives 0.760 in 2.2 h. This small calculation verifies joint assignment logic; an operational fleet schedule must add actual travel and intervention resources.

9 An operating plan tied to its evidence

Preserve independent contact and transmit time-stamped position uncertainty, depth, faults and life-support margins. Deploy from the feasible near-term surviving distribution, update imperfect misses, charge the complete sortie including return, and retain discovery priority for the original time-to-location objective. On the declared nominal ensemble it is faster than both tested references. The optional recovery candidate exchanges 2.35 minutes of restricted mean for 0.823 points of opportunity under a specific bottom-capable reference service; this is a decision tradeoff requiring an actual recovery commitment.

The most consequential preparation condition is a ready, depth-compatible service inside the measured response margin. The seven-hour readiness case removes much more opportunity than route refinement adds. Current error and persistent sonar failure can overwhelm a route that is correct within its own equations. The company should qualify depth and weather limits through fault demonstrations, acoustic trials, equipment readiness and a complete extraction rehearsal, then update the joint model from those measurements. A search probability by itself cannot qualify a depth at which timely passenger recovery is unavailable.

Memorandum to the Government of Greece

To: Ministry responsible for maritime safety, Government of Greece
From: Team 7391856
Subject: Evidence required for MCMS tourist operations
Date: 13 October 2026

We recommend approval only within an operating envelope supported by a demonstrated recovery service. MCMS should define its maximum depth, weather and current limits, tracking quality, crew readiness and response time together. A vehicle that can be found beyond the reach of a timely, compatible recovery system should not be used there for tourist operations.

The host should carry calibrated acoustic tracking and telemetry, sound-speed profiling, navigation records and trained operators. Every submersible should have an independently powered emergency beacon. Before and during each dive, it should send its identity, the measurement time, position estimate and uncertainty, depth and vertical motion, thruster and ballast/leak status, battery and life-support margins. Missed-contact escalation must use the age of the last trusted measurement, including missed intervals and processing delay.

Search deployment should account for three different outcomes: a vehicle resting on the seabed, one suspended in the water, and one still moving after communication fails. A failed survey does not clear an area if shadow or low reflectivity can cause repeated misses. The supervisor should record effort and detection quality, update the remaining location distribution, and follow drifting probability rather than repeating a compact field indefinitely.

In our synthetic deep-Ionian comparison, the recommended discovery-priority plan found the vehicle with 70.9% probability within six hours and 81.3% within eighteen hours. Its restricted mean location time was 6.456 hours. A stronger cumulative-coverage plan with the same reachable field, resources and return reserve gave 70.2% by eighteen hours and 7.726 hours. A compact lawnmower gave 50.2% and 11.411 hours. These are comparative model results under declared conditions. Stronger, rotated currents reduced the recommended plan's final discovery to 47.3%, so local environmental observations and acoustic trials are essential.

The host should carry or secure rapid tracking capability before each dive. Choose purchase, long-term onboard lease or a response contract by complete annual cost and demonstrated readiness; ownership itself is not a safety condition. Quotations must specify maintenance, training, standby, mobilization, vessel and use fees, availability and ready-at-scene delay. For equally capable services with common availability and conditional success $a\rho$, the six-hour rather than seven-hour service is worth an annual premium only up to $0.261729a\rho$ times annual activation frequency times the explicit consequence valuation. This is a conditional threshold, not a market price; measured safety response limits apply regardless of monetary value.

The regulator should require representative fault and missed-contact drills, known-target acoustic trials at both seabed and suspended depths, and rehearsals of the entire response chain. Manufacturer maxima and a promise to mobilize are not evidence that a ready service can complete the task.

Memorandum continued: readiness and emergency action

A published survey AUV example reaches 6,000 metres and an intervention ROV reaches 6,500 metres. Neither rating establishes passenger recovery. A published docking rescue system reaches only 610 metres and needs a compatible hatch; it cannot serve the 3,000-metre scenario. MCMS must demonstrate an engineered emergency ascent or extraction method at each requested depth, including intact hull, ballast release, entrapment, certified rigging and lift points, loads, controlled ascent and surface handling. Passenger transfer also requires pressure and medical compatibility.

The useful response clock includes discovery, confirmation and communication, recovery-service readiness, approach and intervention, ascent, surface handling and a reserve. It begins from the last trusted status and never restarts when another vessel arrives. Our reference calculation assumes eighteen hours of life support, two hours reserved, six-hour recovery readiness and a compatible deep service. Under those assumptions, the recommended search provides a 75.4% timing/depth opportunity. It is not a measured probability of saving passengers.

A recovery-priority search option raises that opportunity to 76.2%, but increases restricted mean location time by 2.35 minutes and lowers final discovery slightly. We therefore retain the faster discovery plan for the original search objective. The alternative is relevant only when a demonstrated bottom-recovery requirement justifies that exchange; its extra opportunity comes from the seabed branch. This choice should be made against actual recovery capabilities rather than assuming that better search automatically means better rescue.

Readiness is more consequential in this example. If the same reference service becomes ready at seven hours rather than six, the recommended route's opportunity falls to 49.2%; readiness at nine hours gives zero within the assumed response deadline, although the search probability is unchanged. MCMS should contract for a ready service at the scene and rehearse that commitment, including adverse weather and maintenance outages. These thresholds are scenario results; the company's measured life-support and engineered service times must replace them before approval.

During an incident the host should preserve the latest telemetry, activate independent emergency acoustics, declare the event using its tested missed-contact rule and call the response team promptly. Search resources should go to the feasible location and depth with the best near-term remaining discovery gain. Repeated misses should prompt current and depth reassessment, equipment-status checks and another sensing modality when necessary. New observations must enter at their actual receipt time without resetting spent endurance.

For several tourist vehicles, require unique identifiers, an acoustic schedule and joint allocation of search and intervention resources. A new destination, including the Caribbean, needs its own environmental information, terrain barriers, launch limits, ports and response agreement. The Greek numerical scenario should not be reused as a foreign-site guarantee.

MCMS should submit trial data, maintenance records, quotations, response contracts and the full extraction demonstration for the requested envelope. Restrict depth or suspend the corresponding operation where that evidence cannot establish a credible response margin. This provides a concrete approval path tied to both location and recovery.

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Report on Use of AI

Identity and role. This is an AI-authored learning revision of an existing development solution to 2024 MCM B. Codex, GPT-6 family, formulated the models, wrote and ran code, interpreted results, selected sources, and wrote/typeset the report and government memorandum. The new episode occurred on 13 October 2026; the earlier scientific solution and corrections occurred on 11 October 2026. No human solving or validation is claimed. The exact service build, tokens and credits are unknown.

Inputs and exposure. The new author read the original four-page problem, the complete current twenty-two-page own manuscript, its original code and scientific evidence, and the generic project methods and lessons. Existing project contact with this same task is retained. This is postseal learning/refinement, not a new unseen-task autonomous test. No new same-task student solution, awarded original, user Downloads or reserved problem was read. General official oceanographic/equipment sources were rechecked at the cited locations.

New work. The AI added a recovery-priority route while preserving time to location as primary, compared common actions/resources, rationally enumerated a three-cell counterexample and separately integrated first-event and reference-recovery deadlines. The planner uses a disclosed private copy of earlier geometry; the validator imports neither planner nor geometry and shares only declared constants, stored trajectories and selected actions. Root independently reconstructed the main results. A separate author-external AI read the complete twenty-page initial paper and original problem, viewed all pages/four figures/two memo pages, and independently checked the eight stored route/environment groups and constraints. It identified unsupported ownership preference and absent realized route/reproduction details. This revision addresses those findings using the existing results, with no new route optimization. The author and root/reviewer are AI roles, not human validation; full transcript and official-policy compliance limitations remain.

Records and failures. The new case retains immutable input hashes, all three numerical attempts, stdout/stderr, actual environment and clocks, first complete manuscript and subsequent version seals. The first validator failed with a syntax error; the second calculated successfully but its evidence objects violated the pipeline's string contract; the third passed the sixteen recorded checks. These records support reproducibility, not reality calibration. Old arrays, failures, sources and frozen manuscripts were retained. The source, all plot arrays, series mapping and current build are bound to the successful run.

Transcript limits. This report and its code, result/check files, source ledger and build/reading records are the available outputs. They are not a complete verbatim prompt/output transcript. Exact historical conversations are not reconstructed as complete, and programmatic evidence is not presented as human approval. The unmeasured current, failure, detection, commercial readiness and engineered-recovery assumptions remain explicit in the scientific report.